

Problem

The feedback resistor of a five-input averaging summing amplifier is 40 kΩ. What are the values of R_1 , R_2 , R_3 , R_4 , and R_5 ?

Step-by-step solution

Step 1 of 6

The output equation of five input summing amplifier is,

$$v_0 = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} + \frac{v_4}{R_4} + \frac{v_5}{R_5} \right)$$

$$v_0 = - \left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 + \frac{R_f}{R_4} v_4 + \frac{R_f}{R_5} v_5 \right) \dots\dots (1)$$

An averaging amplifier is a summer that provides an output equal to the average of the inputs.

The output equation of averaging amplifier with five inputs is,

$$v_0 = -\frac{1}{5} (v_1 + v_2 + v_3 + v_4 + v_5)$$

$$v_0 = - \left(\frac{1}{5} v_1 + \frac{1}{5} v_2 + \frac{1}{5} v_3 + \frac{1}{5} v_4 + \frac{1}{5} v_5 \right) \dots\dots (2)$$

Step 2 of 6

Compare first terms of equations (1) and (2) to get the value of R_1 .

$$\frac{R_f}{R_1} = \frac{1}{5}$$

$$R_1 = 5R_f$$

Substitute 40 kΩ for R_f .

$$R_1 = 5(40 \times 10^3)$$

$$R_1 = 200 \text{ k}\Omega$$

Therefore, the value of R_1 , is 200 kΩ.

Step 3 of 6

Compare second terms of equations (1) and (2) to get the value of R_2 .

$$\frac{R_f}{R_2} = \frac{1}{5}$$

$$R_2 = 5R_f$$

Substitute 40 kΩ for R_f .

$$R_2 = 5(40 \times 10^3)$$

$$R_2 = 200 \text{ k}\Omega$$

Therefore, the value of R_2 , is 200 kΩ.

Step 4 of 6

Compare third terms of equations (1) and (2) to get the value of R_3 .

$$\frac{R_f}{R_3} = \frac{1}{5}$$

$$R_3 = 5R_f$$

Substitute $40\text{ k}\Omega$ for R_f .

$$R_3 = 5(40 \times 10^3)$$

$$R_3 = 200\text{ k}\Omega$$

Therefore, the value of R_3 , is $200\text{ k}\Omega$.

Step 5 of 6

Compare fourth terms of equations (1) and (2) to get the value of R_4 .

$$\frac{R_f}{R_4} = \frac{1}{5}$$

$$R_4 = 5R_f$$

Substitute $40\text{ k}\Omega$ for R_f .

$$R_4 = 5(40 \times 10^3)$$

$$R_4 = 200\text{ k}\Omega$$

Therefore, the value of R_4 , is $200\text{ k}\Omega$.

Step 6 of 6

Compare fifth terms of equations (1) and (2) to get the value of R_5 .

$$\frac{R_f}{R_5} = \frac{1}{5}$$

$$R_5 = 5R_f$$

Substitute $40\text{ k}\Omega$ for R_f .

$$R_5 = 5(40 \times 10^3)$$

$$R_5 = 200\text{ k}\Omega$$

Therefore, the value of R_5 , is $200\text{ k}\Omega$.